Aditya Reddy

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Education

Manipal University Jaipur **Aug 2024 – Expected May 2028**
Bachelor of Technology, Electronics and Communication Engineering *Jaipur, India*

Technical Skills

HDL / RTL Design: SystemVerilog, Verilog

Physical Design: RTL-to-GDS flow, Floorplanning, Placement & Route (PnR), Clock Tree Synthesis (CTS), Static Timing Analysis (STA), DRC/LVS signoff

EDA & Tools: OpenLane 2, OpenROAD, Yosys, Vivado, Magic, KLayout, OpenSTA, SymbiYosys, Verilator, GTKWave, LTspice; SKY130 PDK

Programming: C, C++, Python

Embedded: ESP32, Raspberry Pi, Arduino

Systems: Linux, Git, Make, GCC, RISC-V toolchain, POSIX

Projects

Systolic Array Accelerator – 4×4 Output-Stationary MAC Array **GitHub**
SystemVerilog, OpenLane 2, SKY130B PDK
- Designed a 4×4 systolic array with 16 pipelined Multiply-Accumulate (MAC) processing elements (PEs) and a ping-pong dual-bank register file to overlap BRAM load latency with compute, eliminating inter-matrix stall cycles; validated all 48 FC2 inference tiles cycle-accurate against a golden INT8 MNIST model.
- Ran full RTL-to-GDS on SKY130B via OpenLane 2, diagnosing and resolving hold violations through targeted post-placement buffer insertion; achieved Static Timing Analysis (STA) closure at 76 MHz worst-corner (SS/125 °C) and 137 MHz nominal (TT/27 °C) with zero setup, hold, Design Rule Check (DRC), and Layout vs. Schematic (LVS) violations across all 9 PVT corners; 18 K cells, 0.21 mm^2 core area.

RISC-V Processor Core – RV32IM 5-Stage Pipeline **GitHub**
SystemVerilog, Vivado, Verilator
- Implemented a full 5-stage RV32IM pipeline with EX/MEM forwarding, load-use stall insertion, and 28 SystemVerilog Assertions (SVA) at 100% pass rate across 8 test programs; added single-cycle 64-bit hardware multiply and a 32-cycle non-restoring radix-2 divider that stalls the pipeline via a `ready` signal, covering all M-extension edge cases.
- Built a 3-component dynamic branch predictor – 16-entry Branch Target Buffer (BTB), 256-entry Gshare Pattern History Table (PHT), and 4-deep Return Address Stack (RAS) – synthesised to Artix-7 in Vivado, closing timing at 100 MHz with performance counters quantifying IPC, branch mispredict rate, and stall cycles.

Dual-Clock Asynchronous FIFO – Formally Verified CDC Design **GitHub**
Verilog, OpenRAM, SymbiYosys, cocotb
- Designed a parametric dual-clock FIFO in Verilog with Gray-coded read/write pointers and 2-flop Clock Domain Crossing (CDC) synchronizers; instantiated an OpenRAM pseudo-dual-port SRAM macro (16×64, SKY130) for the storage array, swapped for a combinational stub during formal to bypass non-synthesizable timing constructs.
- Formally verified no-overflow, no-underflow, and CDC safety using SymbiYosys with smtbmc/Boolector — BMC at depth 128 with `multiclock on` for the top-level, and k-induction (`mode prove`) for `rst_sync`, `fifo_mem`, and `sync_fifo` sub-modules; validated with 13 cocotb testcases across `async_fifo` and `fifo_mem`, covering reset state, full/empty conditions, pointer wraparound, concurrent read/write, and asymmetric clock ratios up to 4:1.

USB-C Supply Voltage Monitor
Analog Circuit Design, LTspice
- Designed a multi-threshold LM324 op-amp comparator network to classify USB-C rail voltage into three bands (<4 V, 4–5 V, >5 V) with LED indicators; verified hysteresis-free switching across the full DC sweep in LTspice before confirming on a breadboarded prototype.

Activities

Volunteer — Bison Asha School for Special Needs: supported educational activities and organised inclusive events
2022–2024